

Dinosaur Recalibration: An Exploratory Bradley–Terry Analysis

A Technical Companion to the Interactive Visualizations

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Abstract

We fit Bradley–Terry (BT) power ratings to $\sim 4,494$ crowd-sourced animal-combat matchup entries, classify animals by broad taxonomic and temporal group, and study an explicit *calibrated rescaling scenario* for the interactive visualizations that accompany this note. A graph-connectivity preprocessing step removes components whose scores cannot be placed stably on the common Lion = 100 scale and adds a set of synthetic pseudo-matchups encoding gender, group-size, duplicate-name, and bridge assumptions. These pseudo-observations regularize the graph; they are modeling choices, not independent biological measurements.

The rescaling is anchored by one deliberately strong hypothesis: *Dakotaraptor* is placed above Lion by the same multiplicative score ratio by which Lion is placed above an unarmed Human. This implies a dinosaur factor of approximately $10.14\times$ in the current fitted data. For the visualization, the same factor is extended to all dinosaurs and its square root to other Mesozoic reptiles; those extensions are heuristic and should not be read as biomechanical estimates. Median log–log regressions of rescaled theropod and mammal scores then differ by about $9.2\times$ at 1 kg and, under extrapolation, about $11.9\times$ at 0.1 kg. Because the theropod scores have already been multiplied by the calibration factor, this vertical gap is *not* an independent discovery of a 9–12 $\times$ anatomical advantage: it is a descriptive consequence of the calibrated scenario, combined with the different fitted mass exponents. The analysis is therefore best interpreted as a transparent “what would the ranking look like under this calibration?” exercise rather than an empirical estimate of dinosaur combat performance. All data, code, and interactive visualizations are public.

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1 Introduction

Popular comparisons between dinosaurs and modern animals often lean heavily on body mass. A *Tyrannosaurus rex* weighed in the same broad range as a large elephant, for example, so a mass-centered ranking naturally tends to place them near one another. This note asks a narrower, explicitly counterfactual question: *what would the fitted crowd ranking look like if one imposed a strong anatomy-sensitive calibration instead of leaving the crowd scale untouched?*

The motivation is that body mass is not the only combat-relevant variable. Many predatory theropods carried specialized teeth, jaws, claws, and locomotor adaptations that have no close analogue in a same-mass mammal. The present analysis does not estimate the contribution of those traits from biomechanical data. Instead, it encodes one illustrative calibration involving *Dakotaraptor*, Lion, and Human, propagates that calibration through the ranking, and examines its consequences. The result is therefore a **scenario analysis**, not a measured correction for a demonstrated human bias.

This work serves as a technical companion to the interactive visualizations at <https://github.com/brunchnewton/aethic-code/tree/main/nexic> and is motivated by the Dark Era hypothesis developed in the Nexic reasoning framework [1].

2 Data

2.1 Source

The raw data consists of pairwise animal combat matchups scraped from the Carnivora.net Inter-specific Conflict Directory [7]. Each matchup records two animals, a win probability for the first animal (as inferred from site-user voting), and the number of votes. After scraping, we obtain $\sim 4,494$ matchup entries spanning $\sim 2,369$ unique animals.

2.2 Unicode normalization

The source data contains corrupted Unicode characters (U+FFFD replacement characters from encoding errors). We apply an allowlist-based cleaning procedure: retain only ASCII letters, digits, and common punctuation; replace corrupted characters with a space at word boundaries and delete them within words. A manual corrections dictionary handles the remaining ambiguous cases.

2.3 Filtering

Matchups with reported win probabilities below 5% or above 95% are excluded from the fitted graph. This is a *stability choice*, not a claim that near-unanimous matchups contain no information: extreme proportions can be highly informative, but they also create near-separation and can dominate a finite-score BT fit, especially when vote counts and connectivity are uneven. The 5–95% window therefore trades some genuine signal for a more stable exploratory scale. No threshold-sensitivity analysis is reported here, so results that depend strongly on this truncation should not be treated as robust.

3 Graph Preprocessing

The raw matchup graph has substantial connectivity problems. Some animals appear only on extreme-probability edges, and numerous small clusters are disconnected from the main compo-

ment. In a Bradley–Terry model, disconnected components do not possess a data-determined relative location on a common score scale; regularization can assign them numerical values, but that placement is driven partly by the penalty rather than cross-component match evidence. For example, the House Mouse sits in a small island whose raw fitted location is not meaningfully calibrated to Lion.

We therefore use a preprocessing and augmentation pipeline with several layers of synthetic constraints. These are best understood as *pseudo-matchups*: assumptions introduced to regularize relative placement where the crowd graph is weak or disconnected. They can reduce variance and repair identifiability, but they can also move scores, so downstream results are conditional on these choices.

3.1 Graph cleanup

We exclude three classes of poorly-connected animals:

1. Animals whose *only* matchups have 0% or 100% win probability (283 animals).
2. Animals on two-animal islands in the “real” graph (228 animals across 114 pairs).
3. Animals on larger islands with no plausible bridge to the main component (39 animals across 11 isolated ecosystem clusters: jellyfish, dragonflies, ants, etc.).

This removes 550 animals. The remaining 1,819 animals and 4,087 real matchups form the basis for structural augmentation.

3.2 Gender constraints

For animals with male, female, and base entries in the dataset (e.g. “Cheetah,” “Cheetah (male),” “Cheetah (female)”), we add synthetic matchups that enforce a **geometric mean** constraint: the base entry’s BT score should equal $\sqrt{\pi_{\text{male}} \cdot \pi_{\text{female}}}$.

This is implemented by adding two pseudo-matchups at identical win probability p : “male beats base at p ” and “base beats female at p .” In isolation, those two BT equations place the base at the geometric mean.¹ In the full fit they do not force equality; they pull the solution toward it in competition with the real matchups. We use $p = 0.60$ at pseudo-weight 10. Note that p also encodes a particular male–female separation ($\pi_{\text{male}}/\pi_{\text{female}} = [p/(1-p)]^2 = 2.25$), so the construction is not assumption-free even though the geometric-mean identity itself is independent of p . The current augmentation yields 55 gender constraints across 7 full trios and 41 pairs.

3.3 Group-to-solo scaling

Entries like “Grey Wolf (pack of 5)” are linked to “Grey Wolf” via pseudo-matchups with $P = N^{0.7}/(N^{0.7} + 1)$, where N is the group size. The exponent 0.7 is an explicit heuristic intended to represent diminishing returns from imperfect coordination; it is not estimated from the Carnivora data in this analysis. This produces 270 group-to-solo links at pseudo-weight 3.

3.4 Island bridges

For the remaining disconnected clusters, we add 12 manual bridge pseudo-matchups at weight 3, each connecting an island animal to a comparison animal in the main component with a hand-chosen probability (e.g. House Mouse $\leftrightarrow$ Brown Rat at 20%). These bridges solve an identifiability

¹If $P(\text{male beats base}) = p$, then $\pi_{\text{base}} = \pi_{\text{male}}(1-p)/p$. If $P(\text{base beats female}) = p$, then $\pi_{\text{female}} = \pi_{\text{base}}(1-p)/p$, hence $\pi_{\text{male}}\pi_{\text{female}} = \pi_{\text{base}}^2$.

problem by adding information that was not present in the original graph. Accordingly, absolute placement of formerly isolated clusters is conditional on the bridge choices.

3.5 Subspecies linking

Animals entered under both a generic and a regional name (e.g. “Lion” and “African Lion”) receive a 50% win-probability link at weight 20. Same-species subspecies (e.g. “Tiger” $\leftrightarrow$ “Siberian Tiger”) receive size-adjusted links at weight 10. This pools information across 51 subspecies pairs and regularizes large discrepancies, while still allowing the real matchups to pull linked entries apart:

Pair	Before	After	Fix
Lion / African Lion	1.65×	1.18×	73%
Tiger / Bengal Tiger	3.01×	1.82×	59%
Am. Black Bear / Florida BB	4.85×	1.37×	90%

Table 1: Score ratio between same-species pairs, before and after subspecies linking.

3.6 Summary

The preprocessing adds 388 synthetic matchups to 4,087 retained real ones, producing a connected augmented graph of 1,819 animals. Synthetic matchups use tiered pseudo-weights (3 for bridges/groups, 10 for gender/subspecies, 20 for same-animal duplicates). House Mouse drops from a score of 1,021 on the previous Cheetah = 100 rendering to 3.3 after augmentation—a roughly 300× shift in network placement—while the reported shifts for well-connected animals are below 5%. The first number should not be described as “wrong” and the second as “correct” in an empirical sense: the latter is the score produced by the chosen augmented graph.

4 Bradley–Terry Model

We assign each animal i a latent strength parameter $\pi_i > 0$ such that the probability of i defeating j is:

$$P(i \text{ beats } j) = \frac{\pi_i}{\pi_i + \pi_j} \quad (1)$$

Working in log-space ($\theta_i = \log \pi_i$), the log-likelihood of the observed data is:

$$\ell(\boldsymbol{\theta}) = \sum_{(i,j)} [w_{ij} \log \sigma(\theta_i - \theta_j) + w_{ji} \log \sigma(\theta_j - \theta_i)] - \frac{\lambda}{2} \|\boldsymbol{\theta}\|^2 \quad (2)$$

where $\sigma(x) = 1/(1 + e^{-x})$. For a real matchup with n_{ij} votes and reported fraction p_{ij} favoring i , the fractional counts are $w_{ij} = n_{ij}p_{ij}$ and $w_{ji} = n_{ij}(1 - p_{ij})$; for synthetic pseudo-matchups, the assigned pseudo-weight plays the role of n_{ij} . The penalty $\lambda = 0.01$ provides L_2 regularization. Because BT strengths are identified only up to a common multiplicative scale before penalization and normalization, final scores are subsequently reported relative to the chosen Lion anchor. We optimize (2) using L-BFGS-B.

5 Taxonomy Classification

Each animal is assigned to one of three *rescaling categories*:

1. **Baseline / unscaled**: extant animals, Cenozoic and Paleozoic extinct species, synapsids, and Mesozoic mammals.
2. **Dinosaur**: non-avian Dinosauria from the Mesozoic.
3. **Mesozoic-other**: pterosaurs, marine reptiles, Mesozoic crocodylomorphs, and other non-dinosaurian Mesozoic reptiles represented in the dataset.

The first label is intentionally *not* “non-Mesozoic,” because it contains Mesozoic mammals; it describes the scaling rule rather than geological age. Classification uses manual overrides (~ 400 entries), name-based pattern matching, and a Wikipedia API fallback. The exported category counts reported by the current script are 154 dinosaurs, 55 Mesozoic-other, and 1,603 baseline entries. Those numbers sum to 1,812, seven fewer than the 1,819-node augmented graph reported in Section 3; the difference is a bookkeeping discrepancy in the current pipeline and should be resolved before the counts are treated as final. It does not alter the interpretive point of this note, but it is retained here rather than silently reconciled.

6 Calibrated Anatomy Rescaling

6.1 Lion anchor

We normalize all scores so that Lion = 100. The subspecies linking in preprocessing (§3) ensures that Lion, African Lion, Barbary Lion, etc. share information through the BT model, making a single entry sufficient as the anchor.

6.2 Human baseline

The unarmed human score is taken directly from the “Human” entry, which the gender constraints have already noise-reduced toward the geometric mean of “Human (male)” and “Human (female)”:

$$\text{Human} = 5.15 \quad (\text{Lion} = 100) \tag{3}$$

6.3 The calibration constraint

The central rescaling hypothesis is defined by a single constraint:

$$\frac{\text{Dakotaraptor}_{\text{rescaled}}}{\text{Lion}} = \frac{\text{Lion}}{\text{Human}} \tag{4}$$

In log-space, Dakotaraptor, Lion, and Human are equally spaced. Since Lion = 100 and Human = 5.15:

$$\text{Dakotaraptor}_{\text{rescaled}} = \frac{100^2}{5.15} = 1,940 \tag{5}$$

The corresponding dinosaur scale factor is $f_{\text{dino}} = 1,940/191.3 \approx 10.14$. This factor is calibrated on one predatory theropod and is then applied uniformly to all dinosaur entries in the interactive visualization. That extension—especially to herbivorous and armored dinosaurs—is a heuristic convenience, not something established by the calibration itself. The Mesozoic-other factor $\sqrt{f_{\text{dino}}} \approx 3.18$ is an additional interpolation heuristic chosen to place those taxa between the unscaled baseline and the dinosaur scenario; it has no independent biomechanical derivation.

6.4 Motivation

The Lion–Human comparison is used because it makes the intended kind of anatomical mismatch intuitive: the score gap is meant to represent weaponry, locomotion, and attack mechanics that are not summarized by body mass alone. The *Dakotaraptor* constraint then *asserts*, rather than derives, that an analogous multiplicative mismatch holds one rung higher. This is the creative hypothesis being visualized. A reader who rejects that calibration should reject the resulting rescaled scores rather than treating them as measurements.

6.5 Results

Animal	Crowd	Rescaled	Factor	Category
<i>Tyrannosaurus rex</i>	10,584	107,344	10.14×	dinosaur
<i>Dakotaraptor steini</i>	191	1,940	10.14×	dinosaur
<i>Velociraptor mongoliensis</i>	1.7	16.8	10.14×	dinosaur
<i>Mosasaurus hoffmannii</i>	5,525	17,597	3.18×	mesozoic-other
Lion	100	100	1.00×	baseline
Tiger	148	148	1.00×	baseline
Human	5.2	5.2	1.00×	baseline

Table 2: Selected animals before and after calibrated anatomy rescaling.

7 Weight–Score Analysis

7.1 Median regression

We fit descriptive power laws of the form $\text{score} = Cw^b$ in log-space using median regression,

$$\min_{a,b} \sum_i |\log s_i - a - b \log w_i|.$$

For the 0.5 quantile, the reported pseudo- R^2 is

$$R_{L1}^2 = 1 - \frac{\sum_i |\log s_i - \widehat{\log s_i}|}{\sum_i |\log s_i - \text{median}(\log s)|},$$

equivalent to comparing the fitted absolute-deviation loss with an intercept-only median model. Mammals below 1 kg are excluded from the mammal fit because the matchup graph provides weak cross-scale calibration there. The theropod sample begins at about 0.4 kg. Consequently, predictions below 1 kg for mammals and below 0.4 kg for theropods are extrapolations rather than fits supported by observations at those masses.

No bootstrap or sampling interval is reported for C or b , and species scores themselves inherit uncertainty from crowd composition, graph augmentation, body-mass estimates, and taxonomic matching. The fitted curves should therefore be read descriptively, not as precision estimates of biological scaling laws.

7.2 Results

The fitted power laws are:

$$\text{Theropods: } s = 0.549 \cdot w^{1.263} \quad (\text{pseudo-}R^2 = 0.56, n = 26) \quad (6)$$

$$\text{Mammals } (\geq 1 \text{ kg}): s = 0.060 \cdot w^{1.376} \quad (\text{pseudo-}R^2 = 0.68, n = 35) \quad (7)$$

On the *rescaled* score system, the theropod intercept is about $9.15\times$ the mammal intercept. This gap must not be interpreted as an independent estimate of a body-plan multiplier. The dinosaur rescaling multiplies every theropod score by the same constant and therefore shifts only the fitted log-intercept, not the slope. Dividing the fitted theropod coefficient by $f_{\text{dino}} = 10.14$ gives the corresponding raw-crowd coefficient

$$C_{\text{theropod,raw}} \approx \frac{0.549}{10.14} = 0.0541,$$

which is close to the mammal coefficient 0.060 at 1 kg. Thus most of the post-rescaling vertical separation is intentionally introduced by the calibration. What the regression adds independently is the estimated *slope difference*: the mammal exponent is steeper (1.376 versus 1.263), so the calibrated gap narrows with increasing mass within the fitted model. The fitted curves do not cross over the biologically relevant mass range, but extrapolating either power law far beyond the observed data would not be warranted.

Weight (kg)	Theropod	Mammal	Ratio	Context
0.02	< 0.01	< 0.01	14×	Mouse-sized
0.1	0.03	< 0.01	12×	Rat-sized illustrative mammal
2	1.3	0.2	9×	Squirrel-sized
15	16.8	2.5	7×	Fox-sized (<i>Velociraptor</i>)
75	128	23	6×	Human-sized (<i>Deinonychus</i>)
250	587	119	5×	Lion-sized (<i>Dakotaraptor</i>)
500	1,408	309	5×	Bear-sized (<i>Utahraptor</i>)

Table 3: Predicted scores at equal body mass under the calibrated scenario. Rows below 1 kg rely on extrapolation of the mammal fit, and rows below 0.4 kg would also extrapolate the theropod fit. The changing ratio is a property of the two fitted exponents; its overall level is dominated by the chosen dinosaur rescaling factor.

7.3 Dark Era implications

The fitted curves can be used to generate an intentionally dramatic illustration of the Dark Era hypothesis, provided the numbers are labeled correctly. A 100 g mammal receives a fitted mammal score of about 0.0025. Comparing that fixed baseline with the *theropod regression* at several theropod body masses gives approximate score ratios of $68\times$ at 0.4 kg (the size scale of *Yi qi*), $870\times$ at 3 kg (a *Compsognathus*-scale animal), and $6,700\times$ at 15 kg (a *Velociraptor*-scale animal). These are cross-mass score ratios generated by the calibrated regression, not empirical predator–prey mortality ratios, direct matchup probabilities, or measurements of selection intensity.

The broader qualitative idea—that many Mesozoic mammals were small and that a prolonged nocturnal phase may have reduced antagonistic encounters with dinosaurs—has independent support in the paleobiological literature [4, 5]. Mesozoic mammals were nevertheless ecologically more

diverse than the old stereotype of uniformly tiny, nocturnal insectivores implies, including lineages with larger body sizes and predatory niches [6]. The present score model therefore does *not* establish that dinosaur predation caused the mammalian limbic system, sensory architecture, or later cognitive evolution. Those stronger claims belong, if anywhere, to a separate historical hypothesis requiring independent comparative and fossil evidence.

8 Discussion

The exercise has several layers of uncertainty. The source is a community voting dataset rather than controlled combat data; voters differ in knowledge, interpretation, and matchup assumptions. Bradley–Terry imposes a one-dimensional latent-strength ordering and context-independent pair probabilities, whereas real animal combat depends on environment, age, sex, motivation, weapon use, and matchup-specific tactics. The 5–95% truncation, manual graph bridges, group exponent, gender pseudo-matchups, duplicate-name links, and L_2 regularization all affect the fitted score scale. Extinct-species masses and taxonomic assignments add further uncertainty.

Most importantly, the $10.14\times$ dinosaur factor is not estimated from biomechanics or held-out outcomes. It is the numerical consequence of the Dakotaraptor–Lion–Human calibration, and the square-root factor for other Mesozoic reptiles is an additional visualization choice. Uniformly extending the theropod-derived factor to all non-avian dinosaurs is especially strong: armor, horns, clubs, predatory dentition, and locomotor strategies differ radically among clades. The interleaved ranking in Appendix B should therefore be read as the output of that scenario, not as a validated zoological league table.

What survives these caveats is a more modest result. The augmented BT model provides a reproducible way to turn a messy pairwise-voting graph into a connected relative score system, and the rescaling makes a particular anatomy-first intuition quantitatively inspectable. The median regressions then show how that chosen vertical shift interacts with different fitted mass exponents. In that sense the project succeeds as an exploratory visualization: it converts an informal “dinosaurs may be systematically underrated by mass-centered intuition” hypothesis into an explicit parameter that can be varied, criticized, replaced, or eventually estimated from better data.

A serious empirical follow-up would not ask whether the present calibrated ranking “looks coherent.” It would estimate anatomy effects from independent predictors or outcomes, report uncertainty, test preprocessing and calibration sensitivity, and reserve a held-out set for validation. Until then, the useful status of the $10.14\times$ factor is *hypothesis parameter*, not “correction factor.”

9 Code and Data Availability

All code, data, and interactive visualizations are available at:

<https://github.com/brunchnewton/aethic-code/tree/main/nexic/data>

The pipeline consists of seven scripts:

1. `1_link_scraper.py` — Scrapes matchup URLs from the directory
2. `2_scrape_duels.py` — Extracts pairwise matchup data from each URL
3. `3_preprocess_duels.py` — Graph cleanup and structural constraints
4. `4_power_assembler.py` — Bradley–Terry model fitting
5. `5_categorizer.py` — Taxonomy classification
6. `6_rescale_dinos.py` — Anatomy correction and rescaling
7. `7_curve_fit.py` — Weight–score analysis and median regression

Matchup data are sourced from the Carnivora.net Interspecific Conflict Directory [\[7\]](#).

A Theropod Weight Estimates

Table 4 lists the theropod data points used in the weight–score power law regression (§7), sorted by body mass. Weight estimates are drawn from paleontological literature and are intended as representative adult values. Rescaled scores are the values generated by the calibrated scenario (Lion = 100), not independent biomechanical estimates.

Theropod	Clade	Weight (kg)	Score
<i>Yi qi</i>	Scansoriopterygid	0.4	1.3
<i>Archaeopteryx lithographica</i>	Basal avialan	0.9	0.9
<i>Compsognathus longipes</i>	Compsognathid	3	0.9
<i>Ornitholestes hermanni</i>	Maniraptoran	12	2.8
<i>Linheraptor exquisitus</i>	Dromaeosaurid	12	3.9
<i>Velociraptor mongoliensis</i>	Dromaeosaurid	15	16.8
<i>Dromaeosaurus albertensis</i>	Dromaeosaurid	15	16.4
<i>Coelophysis bauri</i>	Coelophysid	20	5.0
<i>Suskityrannus hazelae</i>	Tyrannosauroid	30	178.2
<i>Stenonychosaurus inequalis</i>	Troodontid	50	45.5
<i>Deinonychus antirrhopus</i>	Dromaeosaurid	73	305.3
<i>Citipati osmolskae</i>	Oviraptorid	75	13.1
<i>Moros intrepidus</i>	Tyrannosauroid	78	88.2
<i>Liliensternus liliensterni</i>	Coelophysoid	130	110.4
<i>Timurlengia euotica</i>	Tyrannosauroid	170	360.4
<i>Ornithomimus edmontonicus</i>	Ornithomimid	170	12.5
<i>Marshosaurus bicentesimus</i>	Megalosauroid	200	877.6
<i>Herrerasaurus ischigualastensis</i>	Herrerasaurid	210	2,529
<i>Dakotaraptor steini</i>	Dromaeosaurid	250	1,940
<i>Achillobator giganticus</i>	Dromaeosaurid	250	1,277
<i>Dilophosaurus wetherilli</i>	Dilophosaurid	400	1,186
<i>Gallimimus bullatus</i>	Ornithomimid	440	662
<i>Utahraptor ostrommaysorum</i>	Dromaeosaurid	500	2,705

Table 4: Theropod data points for the weight–score regression.

B Interleaved Rankings: Mammals and Dinosaurs

Table 5 presents a curated selection of 123 animals from across the rescaled rankings, sorted *descending* by combat score with Lion = 100. Dinosaurs are marked with ▲ and Mesozoic non-dinosaurs with △; unmarked entries are baseline / unscaled. Solo entries only—groups, packs, and female-specific entries are excluded. Weights are adult averages from the paleontological and zoological literature. Familiar reference animals (dog breeds, farm animals, common wildlife) are interspersed throughout as intuitive scale markers.

The interleaving visualizes the consequences of the chosen scaling rule. Under that scenario, a 73 kg *Deinonychus* ties a 450 kg Polar Bear at 305 (rows 40–41), a 250 kg *Dakotaraptor* outscores a 6-tonne Woolly Mammoth (rows 23–24), 15 kg dromaeosaurids rank alongside 55–68 kg mammalian predators (rows 71–74), and a 30 kg *Suskityrannus* scores above Tiger (rows 47 and 52). These juxtapositions are outputs of the calibrated visualization and should not be read as independent validation of the calibration.

#	Score	Animal	Weight ²	
1	114,463	<i>Ankylosaurus magniventris</i>	6.0 t	▲
2	107,344	<i>Tyrannosaurus rex</i>	8.4 t	▲
3	86,822	<i>Giganotosaurus carolinii</i>	6.5 t	▲
4	81,449	<i>Triceratops horridus</i>	9.0 t	▲
5	54,927	<i>Diplodocus carnegii</i>	13 t	▲
6	35,008	<i>Euoplocephalus tutus</i>	2.5 t	▲
7	33,036	<i>Brontosaurus excelsus</i>	15 t	▲
8	29,645	<i>Spinosaurus aegyptiacus</i>	7.0 t	▲
9	28,365	<i>Stegosaurus stenops</i>	3.5 t	▲
10	22,221	<i>Daspletosaurus torosus</i>	2.5 t	▲
11	19,393	<i>Styracosaurus albertensis</i>	2.7 t	▲
12	17,597	<i>Mosasaurus hoffmannii</i>	5.0 t	△
13	17,285	<i>Tarbosaurus bataar</i>	5.0 t	▲
14	14,865	<i>Allosaurus fragilis</i>	2.0 t	▲
15	7,525	<i>Parasaurolophus spp.</i>	2.5 t	▲
16	4,204	<i>Iguanodon bernissartensis</i>	3.5 t	▲
17	3,894	<i>Edmontosaurus annectens</i>	4 t	▲
18	3,883	<i>Carnotaurus sastrei</i>	1.5 t	▲
19	2,981	Columbian Mammoth	10.0 t	
20	2,705	<i>Utahraptor ostrommaysorum</i>	500 kg	▲
21	2,529	<i>Herrerasaurus ischigualastensis</i>	210 kg	▲
22	2,205	Orca (Killer Whale)	5.4 t	
23	1,940	<i>Dakotaraptor steini</i>	250 kg	▲
24	1,447	Woolly Mammoth	6.0 t	
25	1,446	Woolly Rhinoceros	2.5 t	
26	1,277	<i>Achillobator giganticus</i>	250 kg	▲
27	1,274	Megatherium americanum	4.0 t	
28	1,186	<i>Dilophosaurus wetherilli</i>	400 kg	▲
29	985	African Bush Elephant	6.0 t	
30	882	White Rhinoceros	2.3 t	
31	878	<i>Marshosaurus bicentesimus</i>	200 kg	▲

#	Score	Animal	Weight ²	
32	853	Great White Shark	1.1 t	
33	710	<i>Pachycephalosaurus wyomingensis</i>	450 kg	▲
34	662	<i>Gallimimus bullatus</i>	440 kg	▲
35	544	<i>Anzu wyliei</i>	200 kg	▲
36	459	Arctodus simus	800 kg	
37	430	Hippopotamus	1.5 t	
38	417	Saltwater Crocodile	450 kg	
39	360	<i>Timurlengia euotica</i>	170 kg	▲
40	305	Polar Bear	450 kg	
41	305	<i>Deinonychus antirrhopus</i>	73 kg	▲
42	274	Smilodon populator	400 kg	
43	235	American Bison	900 kg	
44	224	<i>Quetzalcoatlus northropi</i>	250 kg	△
45	212	Kodiak Bear	600 kg	
46	203	Smilodon fatalis	220 kg	
47	178	<i>Suskityrannus hazelae</i>	30 kg	▲
48	176	(Wild) Water Buffalo	900 kg	
49	161	Water Buffalo	800 kg	
50	160	Wood Bison	900 kg	
51	150	American Alligator	230 kg	
52	148	Tiger	220 kg	
53	133	Grizzly Bear	270 kg	
54	110	<i>Liliensternus liliensterni</i>	130 kg	▲
55	109	Alaskan Moose	600 kg	
56	100	Lion	190 kg	
57	88	Siberian Tiger	230 kg	
58	88	<i>Moros intrepidus</i>	78 kg	▲
59	85	African Lion	185 kg	
60	81	Bengal Tiger	220 kg	
61	48	Jaguar	96 kg	
62	46	<i>Stenonychosaurus inequalis</i>	50 kg	▲
63	38	<i>Protoceratops andrewsi</i>	180 kg	▲
64	37	Western Gorilla	160 kg	
65	31	Cougar	70 kg	
66	30	Leopard	60 kg	
67	24	Eastern Gorilla	160 kg	
68	23	Komodo Dragon	70 kg	
69	20	Green Anaconda	30 kg	
70	18	Wild Boar	80 kg	
71	17	Spotted Hyena	55 kg	
72	17	Dire Wolf	68 kg	
73	17	<i>Velociraptor mongoliensis</i>	15 kg	▲
74	16	<i>Dromaeosaurus albertensis</i>	15 kg	▲
75	13	<i>Citipati osmolskae</i>	75 kg	▲
76	13	Elk	330 kg	
77	13	<i>Ornithomimus edmontonicus</i>	170 kg	▲

#	Score	Animal	Weight ²	
78	10	Common Warthog	80 kg	
79	9.7	Grey Wolf	40 kg	
80	7.9	Burmese Python	90 kg	
81	7.5	Neanderthal	78 kg	
82	6.4	Bornean Orangutan	75 kg	
83	5.9	Human (male)	80 kg	
84	5.9	Neapolitan Mastiff	60 kg	
85	5.8	Ostrich	110 kg	
86	5.8	<i>Psittacosaurus lujiatunensis</i>	20 kg	▲
87	5.7	Dogo Argentino	43 kg	
88	5.2	Human	70 kg	
89	5.0	<i>Coelophysis bauri</i>	20 kg	▲
90	4.8	Caucasian Shepherd Dog	55 kg	
91	4.5	<i>Pteranodon longiceps</i>	25 kg	△
92	4.5	Rottweiler	50 kg	
93	4.4	Cheetah	50 kg	
94	4.3	Tibetan Mastiff	55 kg	
95	4.2	Red Kangaroo	85 kg	
96	3.9	<i>Linheraptor exquisitus</i>	12 kg	▲
97	3.9	Wolverine	14 kg	
98	3.5	Mandrill	25 kg	
99	3.3	Common Chimpanzee	50 kg	
100	3.1	Kangal	55 kg	
101	2.9	Southern Cassowary	60 kg	
102	2.8	<i>Ornitholestes hermanni</i>	12 kg	▲
103	2.5	Boerboel	65 kg	
104	2.2	Thylacine	25 kg	
105	2.1	German Shepherd	34 kg	
106	1.8	Human (female)	55 kg	
107	1.8	Ocelot	12 kg	
108	1.7	King Cobra	6 kg	
109	1.7	American Pit Bull Terrier	27 kg	
110	1.7	Honey Badger	11 kg	
111	1.5	Bonobo (Pygmy Chimpanzee)	40 kg	
112	1.5	Caracal	13 kg	
113	1.4	Fishing Cat	10 kg	
114	1.3	Bobcat	9 kg	
115	1.3	<i>Yi qi</i>	400 g	▲
116	1.2	Coyote	14 kg	
117	1.1	Labrador Retriever	30 kg	
118	1.0	Savannah Cat	8 kg	
119	0.9	<i>Archaeopteryx lithographica</i>	900 g	▲
120	0.9	<i>Compsognathus longipes</i>	3 kg	▲
121	0.5	Brown (Norway) Rat	300 g	
122	0.1	House Mouse	20 g	

#	Score	Animal	Weight ²
123	0.1	Etruscan Shrew	2 g

Table 5: Interleaved scores under the calibrated rescaling scenario, descending (Lion = 100, row 56). $\blacktriangle$ = dinosaur, $\triangle$ = Mesozoic non-dinosaur. Dog breeds, farm animals, and common wildlife are interspersed as intuitive scale markers. Key juxtapositions: rows 40–41 (73 kg *Deinonychus* = 450 kg Polar Bear), rows 23–24 (250 kg *Dakotaraptor* > 6 t Woolly Mammoth), rows 71–74 (15 kg raptors $\approx$ 55–68 kg wolf-tier predators).

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